ForgeData™

The Data Layer That Ships Where the Mission Runs

ForgeData™ is the data-intelligence layer of the Forge suite: it decides which geospatial dataset serves a request and how to read it, across a dozen incompatible sources, and returns a structured, machine-readable reason rather than a plausible-but-wrong dataset when nothing qualifies. Its catalog is a single local file with no secrets inside, so a curated catalog and the data it references travel together as a bundle to a field laptop or a disconnected site and answer on arrival. One JVM process, local disk, bundled with the suite.

One file

a portable catalog with no secrets

one GeoPackage holds datasets and source recipes; keys stay in the environment, never in the file

Local

no required external service

in the suite it runs as a child process of the agent over stdio, reading files on local disk

One JAR

nothing to install

forgedata-1.2.6-all.jar: Java 17+, no daemon, no native install step; the suite brings its own runtime

Refuses, never fabricates

Failures return a structured, machine-readable reason — an area out of coverage, a malformed bounding box, an unsupported read, a catalog that is read-only — rather than a confident-looking wrong answer; since 1.2.6 every reply with no data names why. When nothing covers an area, it names the registered source whose sync could acquire it. The data layer fails closed, the same discipline the rest of the suite is built on.

Honest terrain, one seam

SRTM flattens open water to a literal 0 m that reads as solid ground. Given a water-mask layer, ForgeData recodes over-water cells to NoData at its elevation read seam, so elevation answers over a coastline are honest — whichever source they come from. Since 1.2.6 every answer also states its vertical units and datum, and projected DEMs read correctly.

Provenance you can audit

Rows ForgeData adds itself are stamped with how and when they arrived — sync, upload, derived product, recording, scenario; published files carry a SHA-256 content hash and can be re-verified in bulk to catch corruption or substitution. Derived products record the operation and parameters that made them, and their parent when it is in the same catalog, so the trail mission work requires is recorded in the catalog, not reconstructed after the fact.

Hardened in 1.2.6. A testing and correctness release: distances and areas on the WGS84 ellipsoid, core queries checked against GDAL, credentials masked in every log line and error reply, and the PostgreSQL driver updated for two security fixes.

Answers that state their reference. Since 1.2.6 every elevation answer carries its vertical units and datum, and says whether they came from the file or from the product's documentation — SRTM on EGM96, Copernicus on EGM2008, 3DEP on NAVD88 over the conterminous US. Values are reported as stored, never converted, and never guessed: an unstated reference reads `unknown`.

CONTINUED ON REVERSE →

BUILT FOR THE WAY MISSIONS HANDLE DATA

Routing across the real, mixed estate

Mission data is never in one place: terrain from an SRTM mirror or USGS 3DEP, imagery from a STAC catalog, hydrography from an internal PostGIS, features from an ArcGIS service, bathymetry from NOAA, point clouds from local LAZ. ForgeData makes all of it addressable by intent and returns the best-fit correct dataset, ranked by coverage and resolution with access cost only breaking the tie.

Ships as a data pack

The catalog and its source recipes live in one GeoPackage with no secrets inside, so a curated bundle is a single artifact a program drops onto an edge device or disconnected host and queries immediately — readable in QGIS or DB Browser for independent inspection.

One curated catalog per workstation

Since 1.2.5: when the suite is installed for everyone on a workstation, each operator keeps a private catalog and reads a shared, read-only one for the whole machine. Publishing into it takes Administrator approval, and writing to it from an account returns a structured refusal — so what every operator sees is controlled.

DEPLOYMENT POSTURE

Secret-free

the catalog stores only the name of the variable that holds a key; in the suite the keys live in the operating system's credential store

Sandboxed

full geometry results are written only under an export root — parent paths refused, symlinks re-checked

Bounded

dataset lists page by default and large geometry is not inlined, so a default request stays sized for the agent that asked it

> **Fail-closed on coverage.** When no dataset covers a requested area, ForgeData does not silently return the nearest tile or an empty success. It returns a structured reason — and, for coverage gaps, the registered source that could acquire the area. The operator learns what is missing instead of acting on a wrong product.

> **Correct in any projection.** A DEM in UTM, a LiDAR-derived elevation product or a WCS coverage in its native CRS used to be read as if its metres were degrees, and answered no coverage without an error. Since 1.2.6 each is read in its own coordinates; LiDAR clouds in US survey feet are converted, and a cloud whose units cannot be known is refused.

Request the Technical Brief and an evaluation build

The Technical Brief covers the catalog, the cost model, every source type, the MCP surface and deployment. ForgeData is bundled with the Forge suite and is not sold separately. The Forge suite is commercially licensed; evaluation builds are available to qualified programs.

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